

Zemax

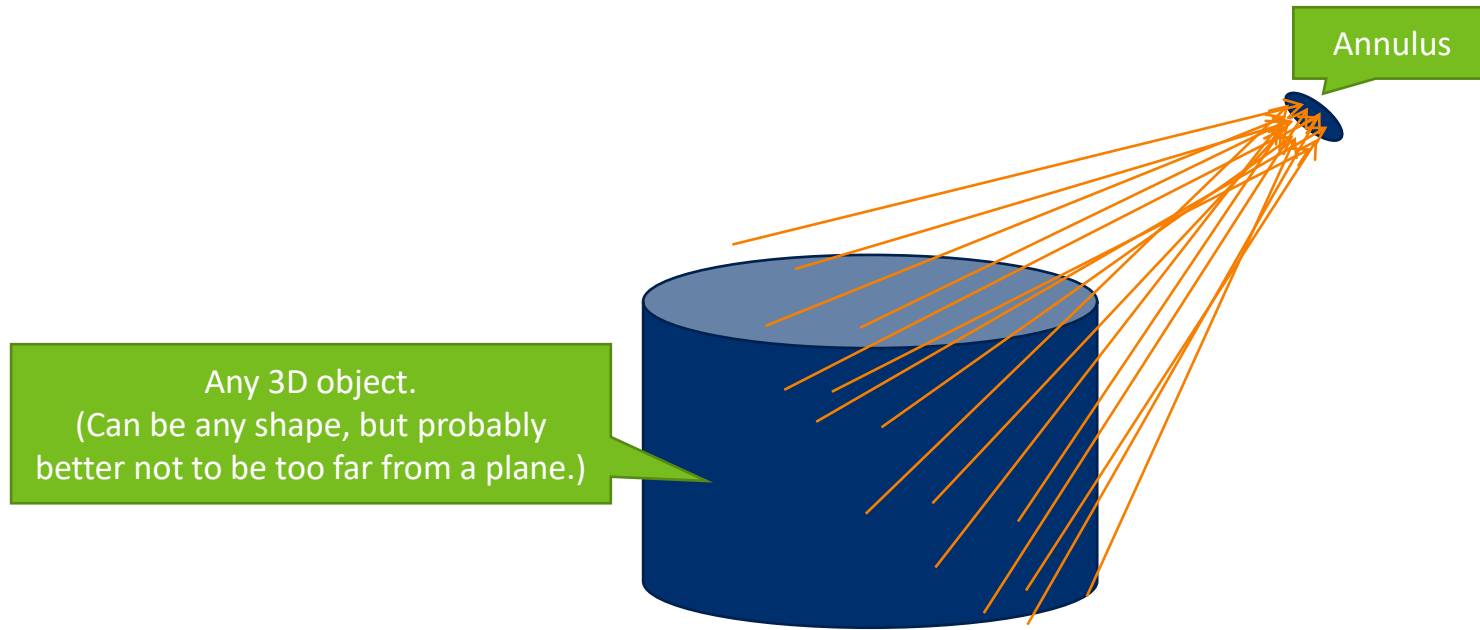
Radiance Camera

Content

- This document explains the required math and optics in the source code.

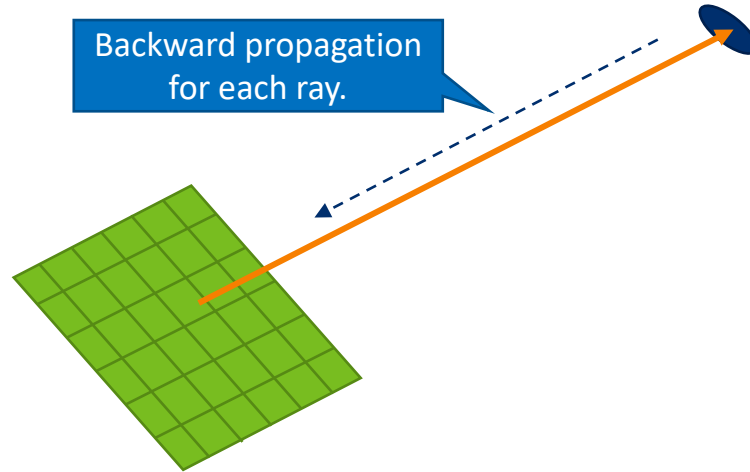
Save rays

- The first step is to save rays hitting the Annulus into a SDF or DAT source file.



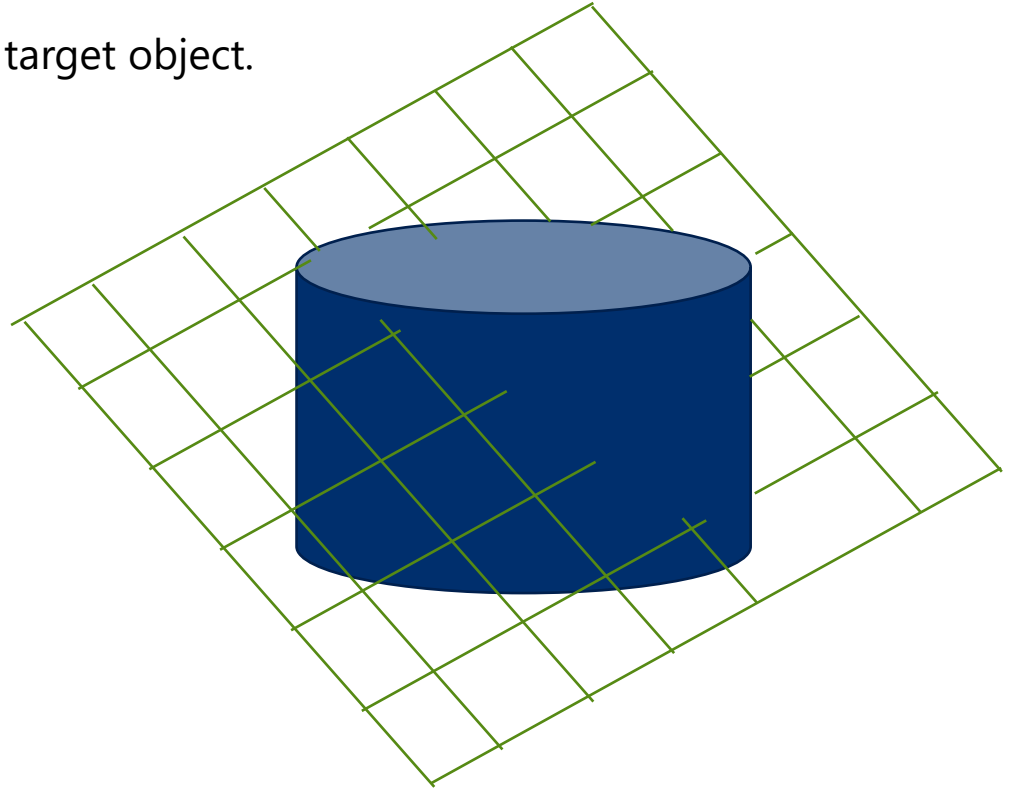
Back propagation

- In API, we read the source file and back propagate these rays to a defined (virtual) plane, count flux on each pixel, and then get irradiance $E = \text{flux}/\text{area}$ of each pixel. The calculation of irradiance is just like what we do for detector in NSQ mode.



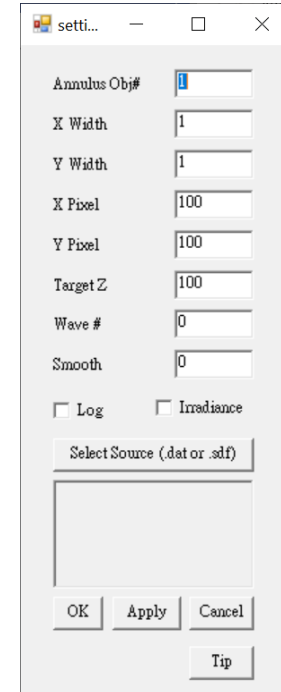
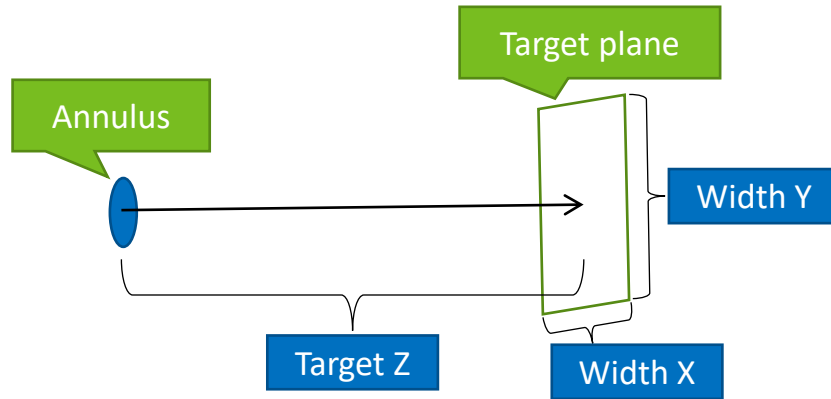
Target plane

- The plane is virtual.
- It's supposed to be overlapping on the target object.



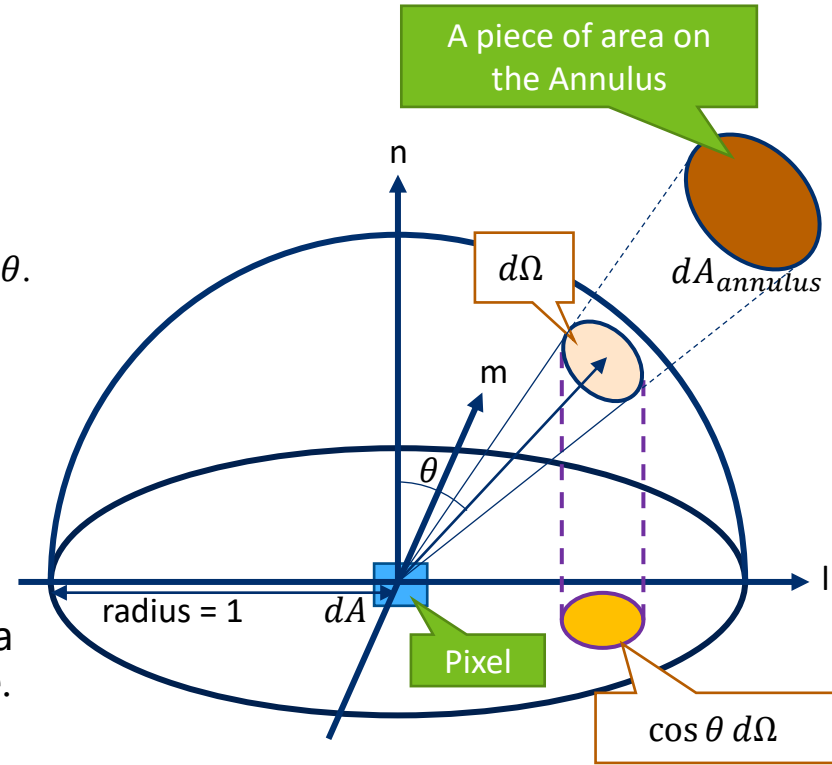
Parameters

- The target plane's position is relative to the Annulus.
- Direction of width X and Y are also based on the Annulus' local coordinate.



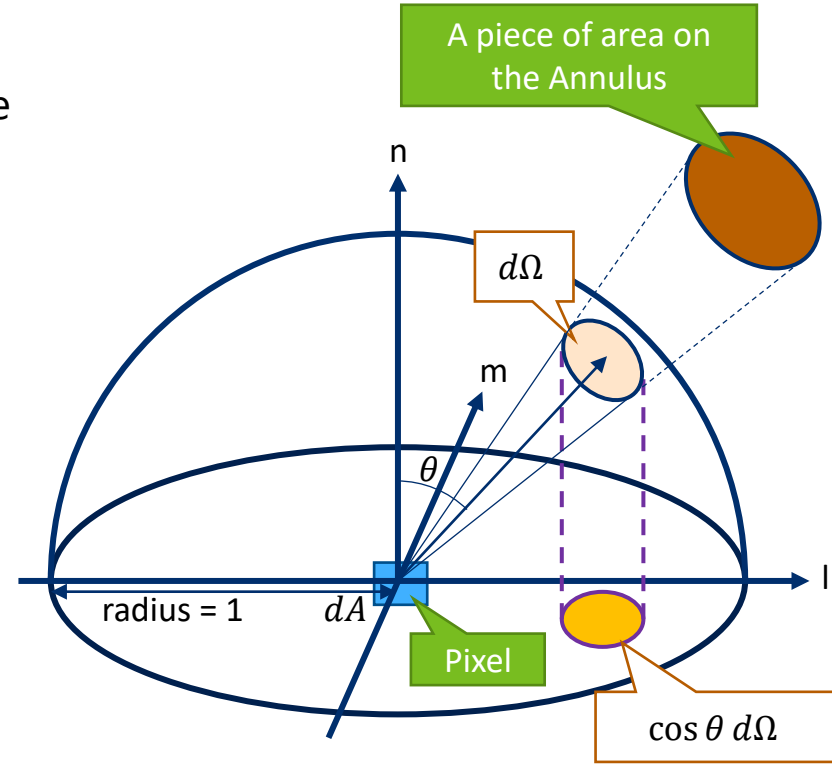
Get radiance

- First look at the equation of radiance.
- $$Radiance = L = \frac{d^2\Phi}{d\Omega dA \cos\theta} = \frac{d\left(\frac{d\Phi}{dA}\right)}{d\Omega \cos\theta} = \frac{dE}{d\Omega \cos\theta}$$
 - Radiance is calculated by dividing irradiance by $d\Omega \cos\theta$.
 - $d\Omega \cos\theta$ is called projected solid angle (PSA).
- Solid angle (SA, $d\Omega$) is same as the area projected from the Annulus (or any object with any shape) on the spherical shell with radius = 1.
 - Any point on this sphere corresponds to a ray propagation direction (l,m,n)!
- And projected solid angle (PSA, $\cos\theta d\Omega$) is the area further projected on the bottom of this hemisphere.
 - This projected plane is basically the LM space!



Get radiance 2

- Here we assume the radiance is uniform in the cone from each pixel to the Annulus. Then we have...
- $L = \frac{dE}{d\Omega \cos\theta}$
- $\Rightarrow L \int d\Omega \cos\theta = \int dE$
- $\Rightarrow L \times PSA_{annulus} = E$
- $\Rightarrow L = \frac{E}{PSA_{annulus}}$
- Radiance is basically irradiance divided by the integrated PSA of the Annulus to each pixel.



Calculating PSA

- The integration $\int d\Omega \cos\theta$ is actually not simple.
- Each small area on the Annulus has different area in LM space.
- The equation cannot be solved analytically.
- We actually didn't use this equation in the code.

$$dA = r dr d\theta$$

$$(r \cos \theta, r \sin \theta)$$

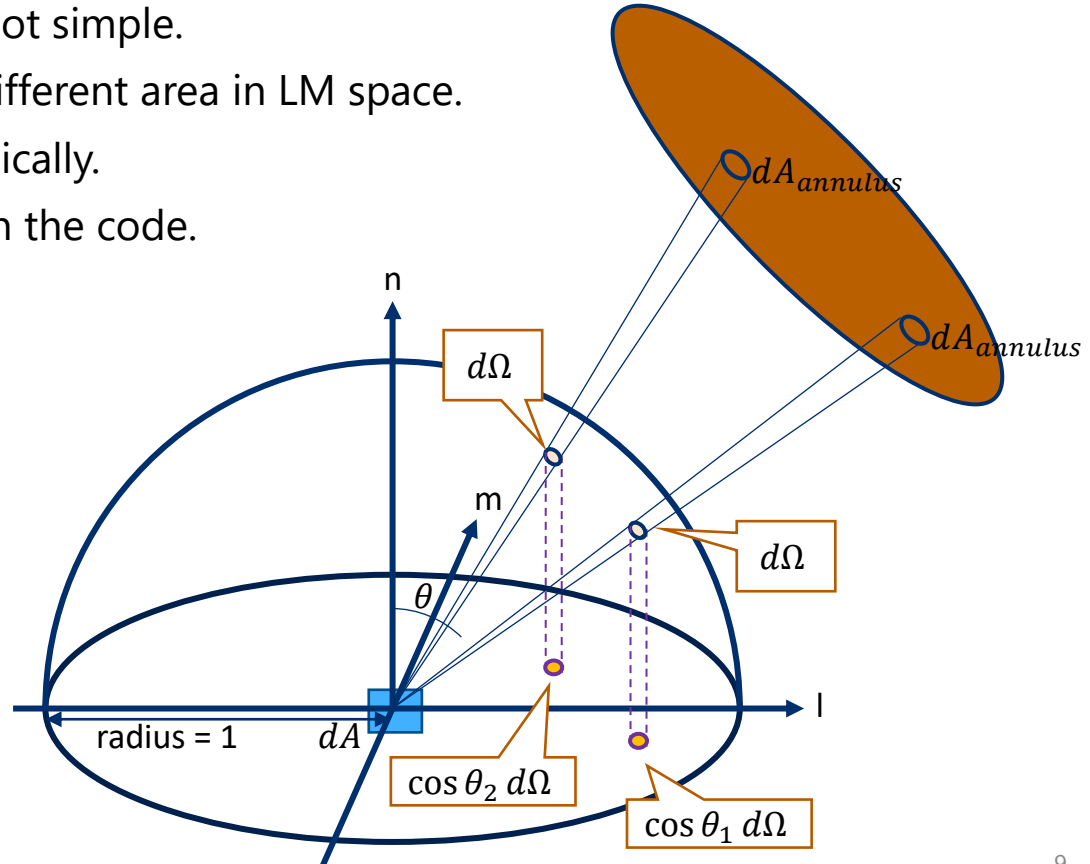
$$D^2 = (x_0 - r \cos \theta)^2 + (y_0 - r \sin \theta)^2 + h^2$$

$$\cos \theta = \frac{h}{D}$$

$$d(SA) = \frac{dA \cos \theta}{h^2}$$

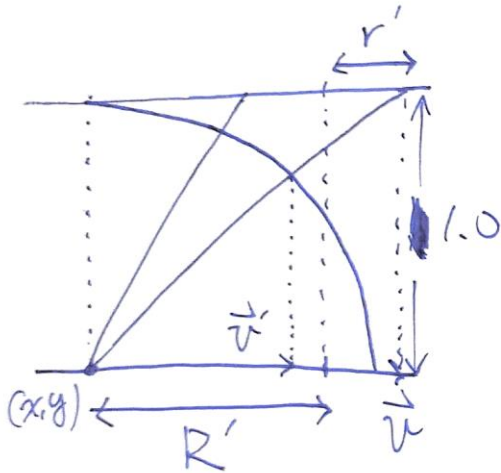
$$d(PSA) = d(SA) \times \cos \theta = \frac{dA \cos^2 \theta}{h^2}$$

$$PSA = \int_0^{R/h} \int_0^{2\pi} \frac{h^2 r dr d\theta}{[h^2 + (x_0 - r \cos \theta)^2 + (y_0 - r \sin \theta)^2]^2}$$



Calculating PSA 2

- Instead, the trick is to get the outline projected from the Annulus onto LM plane first.
 - h = distance from Annulus to the pixel in "z" direction.
 - R = radial distance from the pixel to Annulus. R' is normalized by h .
 - r = radius of Annulus. r' is normalized by h .
 - vector $\vec{v}$ is the coordinate of outline on Annulus.
 - vector $\vec{v}'$ is the projection of $\vec{v}$ on the LM plane.



$$R' = \frac{R}{h}, \quad r' = \frac{r}{h}$$

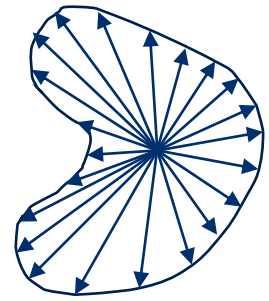
$$|\vec{v}|^2 = R'^2 + r'^2 + 2R'r'\cos\phi$$

$$\vec{v} = (R', 0) + r'(\cos\phi, \sin\phi)$$

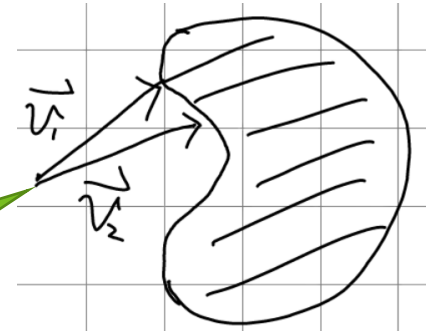
$$\vec{v}' = \frac{\vec{v}}{\sqrt{|\vec{v}|^2 + 1}} = \frac{(R' + r'\cos\phi, r'\sin\phi)}{\sqrt{R'^2 + r'^2 + 1 + 2R'r'\cos\phi}}$$

Calculating PSA 3

- Usually, the projected area in LM plane is something like right picture.
 - Like a "pea"?
- Now the good question to ask is how to calculate the area if I have equation of the outline.
- The method we use is to split it into many triangles and use determinant to calculate the area of each triangle.
- Then the task is simply to
 - sample the outline by many points.
 - build vectors from an arbitrary point to each point on the outline
 - calculate and sum the area of triangles formed by each of neighbor triangles
- Note it's no problem if you select a point that is outside of the outline to draw vectors. (guess why :D)



A diagram on a grid showing a triangle. Two vectors, $\vec{v}_1$ and $\vec{v}_2$, originate from the same vertex of the triangle. $\vec{v}_1$ points to the top vertex, and $\vec{v}_2$ points to the right vertex. The triangle is shaded with diagonal lines.
$$A = \left| \begin{vmatrix} v_{1x} & v_{1y} \\ v_{2x} & v_{2y} \end{vmatrix} \right| \times 0.5$$



Outside is OK